

Jinkyounghwangbo

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EDUCATION

University of Illinois Urbana-Champaign

Master of Science in Computer Science

- Advised by Prof. Minje Kim

Aug 2026– May 2028 (Expected)

Illinois, USA

Seoul National University

Bachelor of Science in Industrial Engineering and Artificial Intelligence

- Honor: Summa Cum Laude, GPA: 4.06/4.30, Rank: 3rd

Mar 2018–Aug 2022

Seoul, Republic of Korea

Sejong Academy of Science and Arts

National STEM-gifted high school

Mar 2015–Jan 2018

Sejong, Republic of Korea

WORK EXPERIENCE

Samsung Electronics

Full Time Researcher @ Samsung Research, Mobile eXperience Business

- Audio Eraser for Galaxy S Series 

Jul 2022–Jun 2026

Seoul, Republic of Korea

- Designed an on-device AI sound detection model working in real time, improving accuracy by over 15% while cutting computing cost by 10%
- Formulated a method that analyzes both audio and video to automatically identify meaningful highlight moments, resulting in two patent filings [P.1, P.2]
- Conducted research on audio-visual separation, using visual cues to isolate individual sound sources
- Acoustic Anomaly Detection for Samsung Home Appliances
 - Designed a training method that resolves subjective labeling, achieving 96.3% fault-detection accuracy – outperforming human experts (87.3%) in blind testing
 - Devised a data-efficient algorithm using <4 hours of training data, generating \$1.3M in estimated annual cost reduction and selected for demonstration at Samsung Tech Fair 2024

Samsung Electronics

Research Intern @ Samsung Research

- Speech Bandwidth Extension with Hifi-GAN

Summer 2021

Seoul, Republic of Korea

- Created a generative model extending speech audio bandwidth from 8 kHz to 16 kHz, improving perceptual quality scores by 31% across diverse speakers and conditions

Seoul National University

Research Intern @ Human Centered Computing Lab (Advisor: Prof. Bongwon Suh)

- ECG-based Heart Disease Prediction

Summer 2020

Gwanggyo, Republic of Korea

PATENTS

[P.2] Jinkyounghwangbo, Hosang Sung, Seong-Hu Kim, et al. "Electronic Device and Method for Summarizing Video Data". KR Patent Application, P2025-0128738. Filed: Sep 10, 2025.

[P.1] JeongHoon Lee, Yeaseul Song, Jinkyounghwangbo, et al. "Electronic Device and Method for Generating Summarized Video Data". KR Patent Application, P2025-0115037. Filed: Aug 19, 2025.

HONORS AND AWARDS

- **Samsung Research Award (\$2,200 USD)** Dec 2024
Biannual award for full-time research-oriented employees; given for work on the Audio Eraser feature.
- **Merit-based Scholarship (Top 5%) @ Seoul National University** 2019–2022
Full amount (3 semesters), 50% (3 semesters), and 10% (1 semester) of tuition waiver.
- **Reporter Award @ Seoul National University** Nov 2019
Awarded by the university president to student reporters; recognized for work as a photojournalist for the university newspaper.

TEACHING / MENTORING EXPERIENCE

- **Teaching Assistant** Mar 2021–Jun 2022
Courses: Human Factors Engineering (406.304), Scientific Management (406.211)
- **Volunteer Social Service**
 - SNU Mentoring Program: Served online 1:1 mentoring for two high school students. Mar–Nov 2021
 - Vocational training TA for young adults with developmental disabilities at a community welfare center. Spring 2019

ACTIVITIES

- **Tobigs: Big Data and Machine Learning Study Club** [🌐] Jan 2021–Jan 2022
Cross-university club where students study machine learning together and build team projects.
 - Smootify [🔗]: Developed system for seamless song transitions, enabling continuous infinite streaming using music generation techniques with MuseGAN.
 - MPTI [🔗]: Developed chat service that allows users to interact with movie characters using DialoGPT and GPT-2.
- **Seoul National University News** [🌐] Apr 2018–Dec 2019
 - Head of Photography Department – Led the department as a photojournalist for the university newspaper.

SKILLS

- **Tools and Frameworks:** PyTorch, TensorFlow, Tensorflow Lite, NumPy, Pandas, Matplotlib, SciPy, scikit-learn
- **Programming Languages:** Python, C/C++, Java, Kotlin, MySQL
- **Others:** Git, Github, Unit Testing, OOP, CI/CD
- **Languages:** Korean (Native), English (Fluent; TOEFL iBT 5.5/6.0)